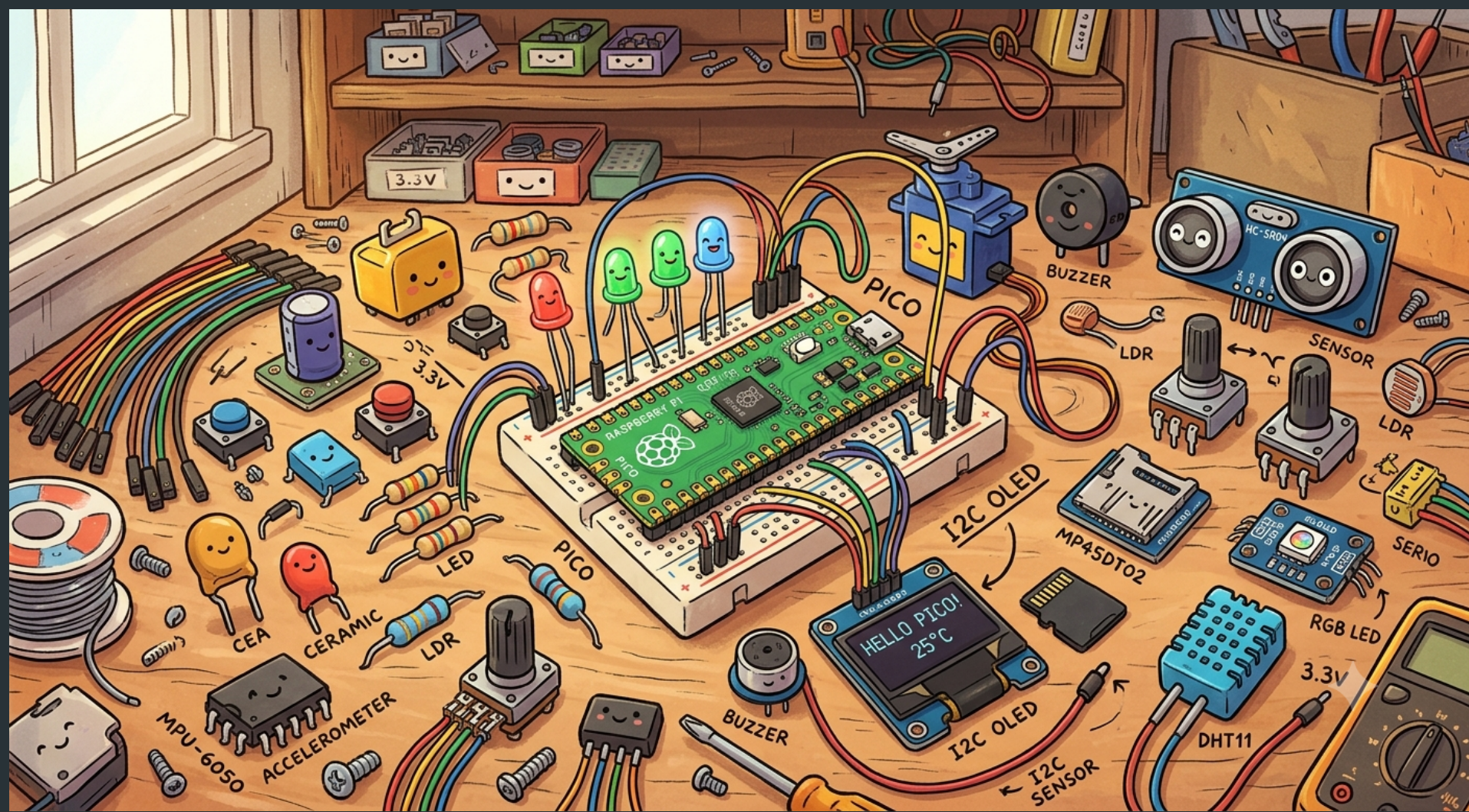




generated with Nano Banana 2

kokchun giang

edge computing - different types of hardware components

we'll **focus** on a few basic hardware components to get started



breadboard



push button



capacitor



ultrasonic sensor



LED



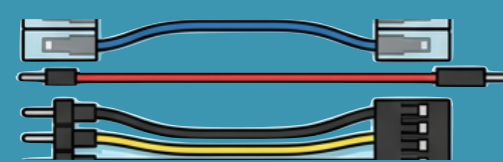
piezo buzzer



servo motor



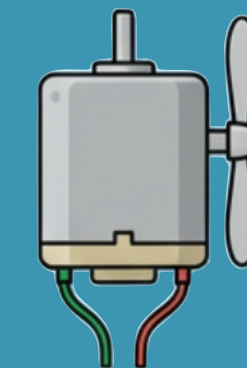
IR sensor



M-M, M-F, F-F
jumper wires



potentiometer



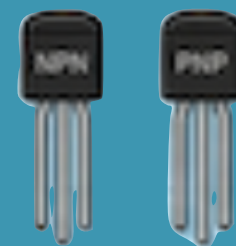
DC motor



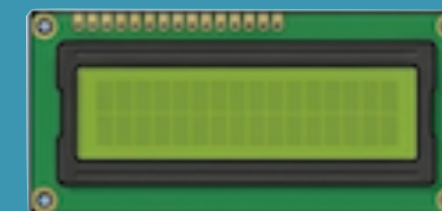
DHT sensor



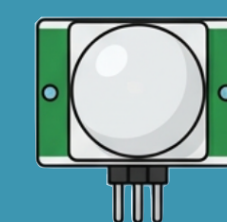
resistor



transistor



display



PIR sensor

we'll **focus** on a few basic hardware components to get started



breadboard



push button



capacitor



ultrasonic sensor



LED



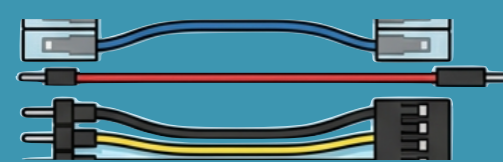
piezo buzzer



servo motor



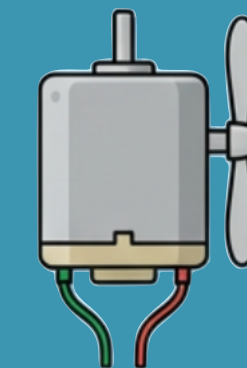
IR sensor



M-M, M-F, F-F
jumper wires



potentiometer



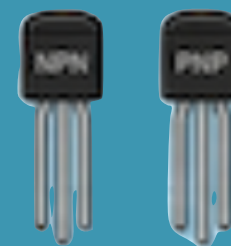
DC motor



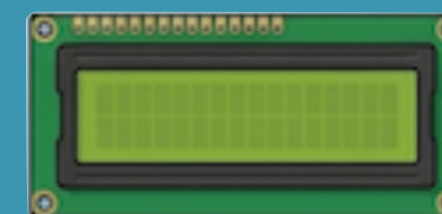
DHT sensor



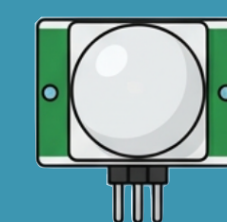
resistor



transistor



display



PIR sensor

a mental **grouping** of these components

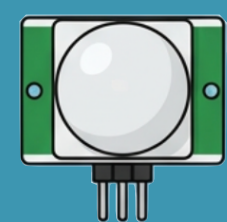
inputs



IR sensor



potentiometer



PIR sensor



DHT sensor

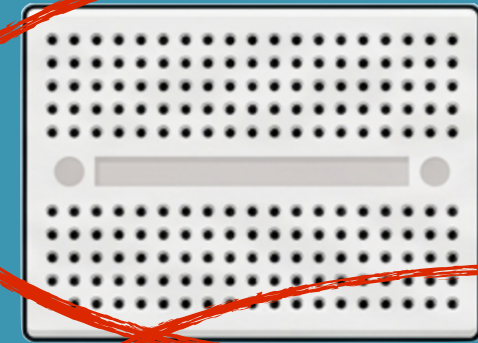


ultrasonic sensor

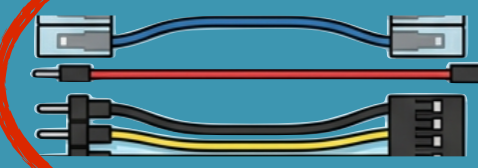


push button

circuit construction



breadboard

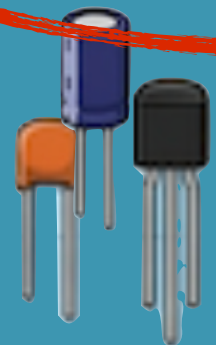


M-M, M-F, F-F jumper wires

electrical control



resistor

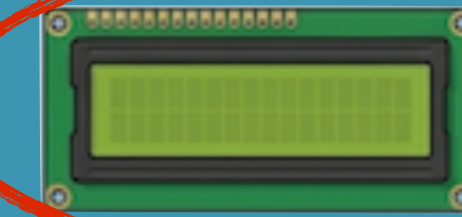


capacitor



transistor

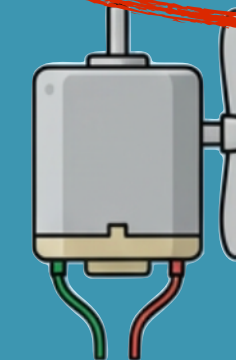
outputs/actuators



display



servo motor



DC motor

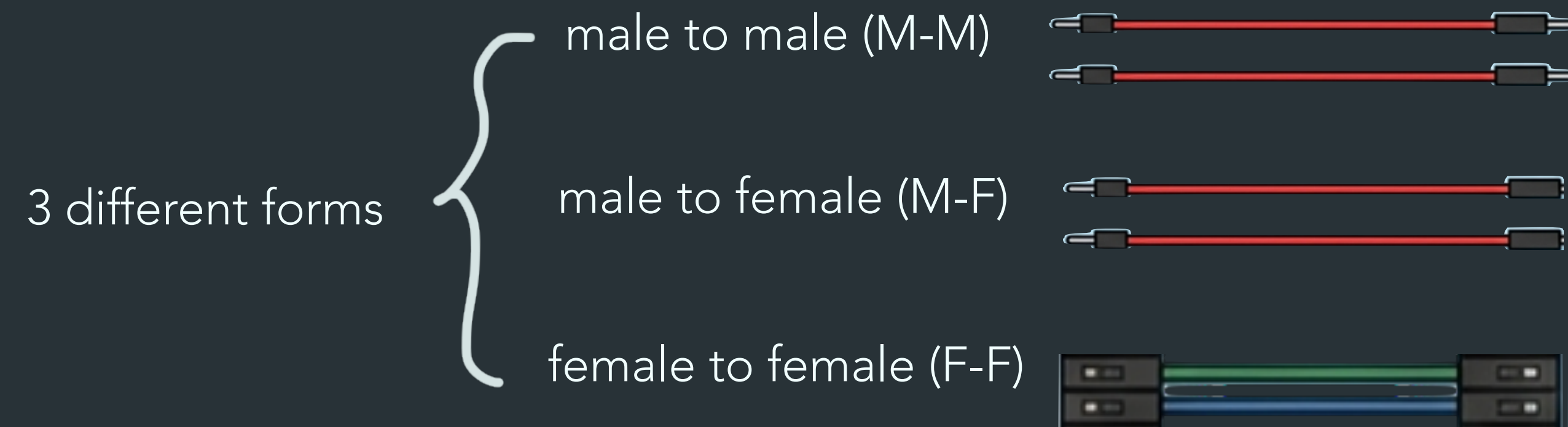


LED



piezo buzzer

connect components with **jumper wires**



limit the current with **resistor**



“acts as narrow
pipe in water
system”

multimeter to measure

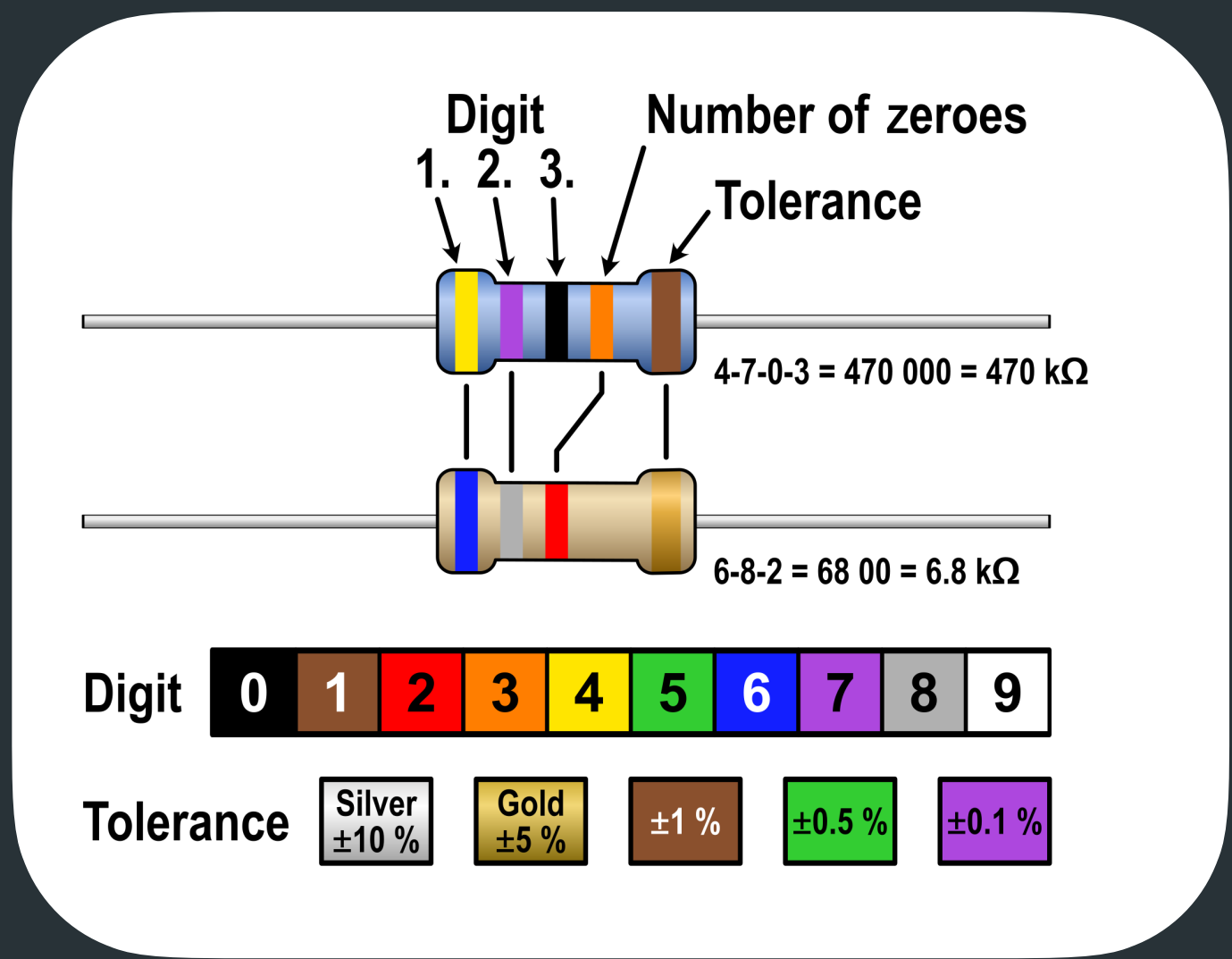


color coding to read resistance

↑ resistance → ↓ current

Ohm's law

$$I = \frac{V}{R}$$



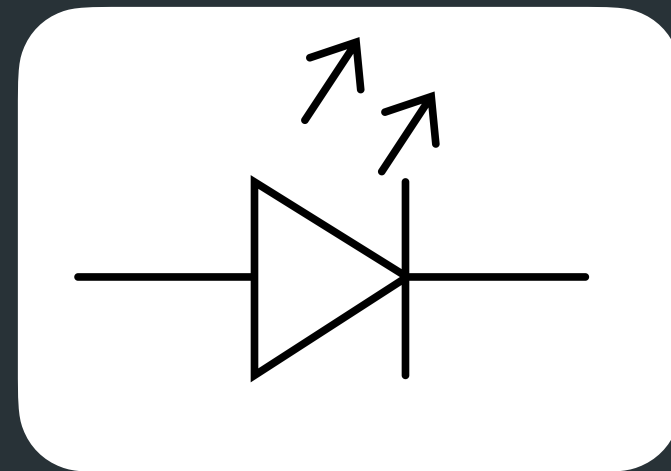
Light emitting diode - LED



longer leg (anode, +) of LED
connect to GPIO through resistor

shorter leg (cathode, -) connected to
GND

current flows
one direction



schematic

LED connected
directly to GPIO

LED draws too
much current

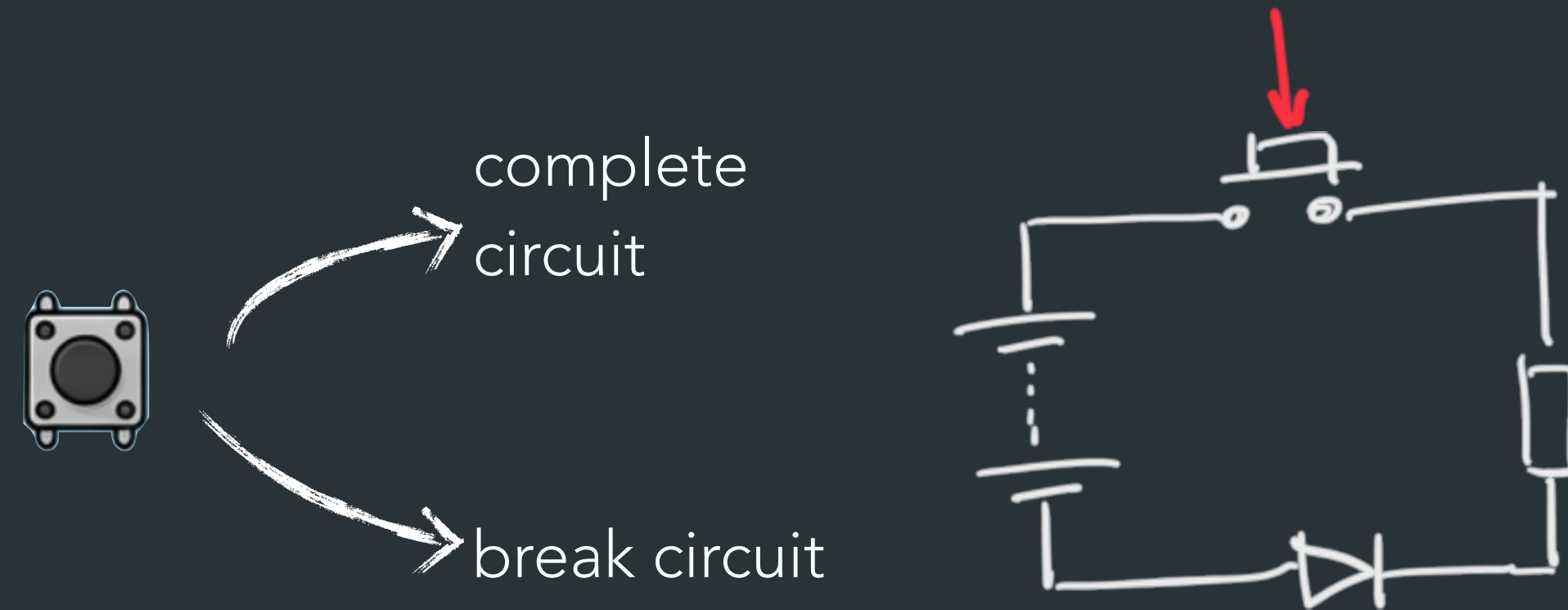
damage LED

damage GPIO

connect a resistor $\geq 330 \Omega$
between LED and GPIO

$\uparrow$ resistance $\rightarrow$ $\downarrow$ light

a simple switch - **push button**



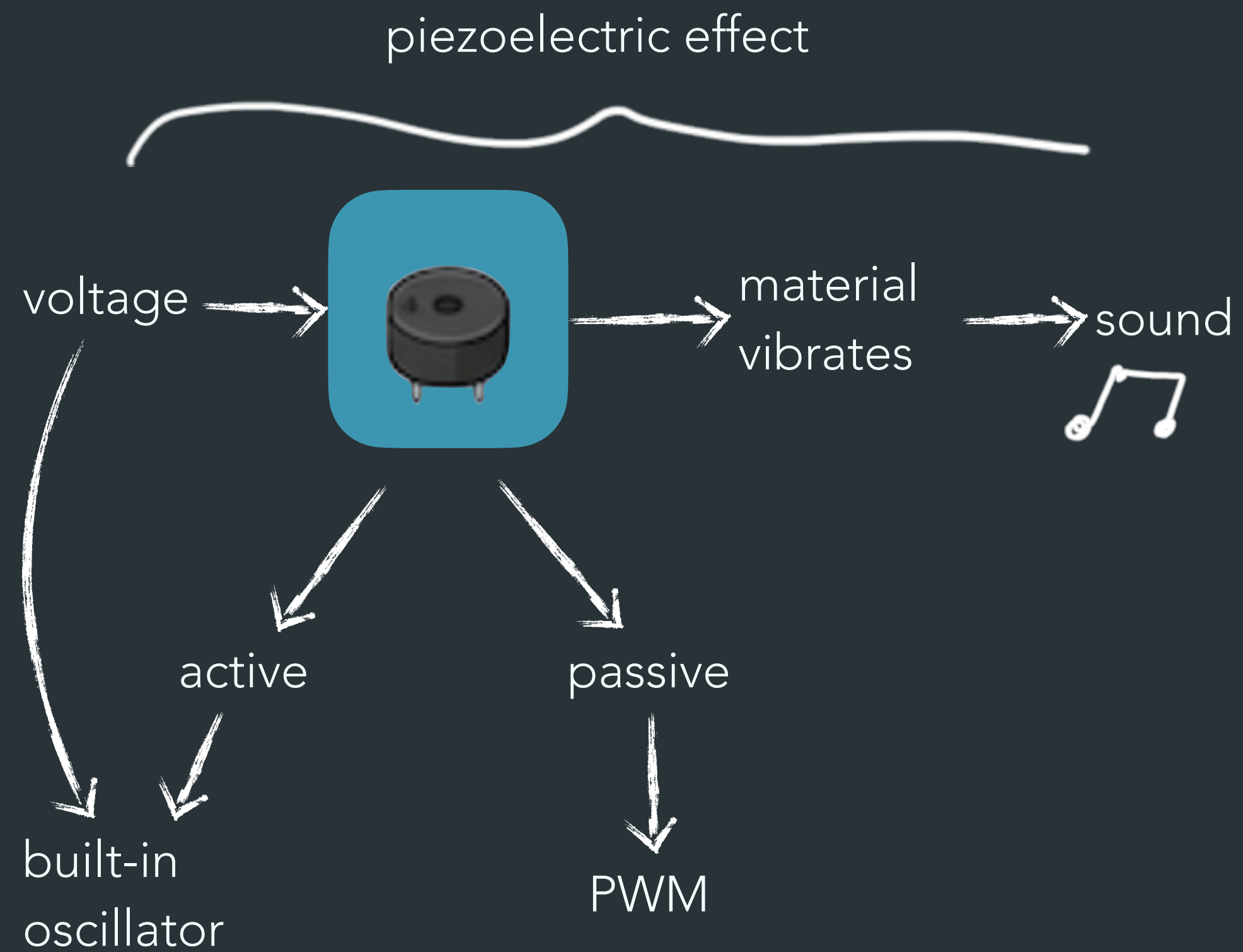
use cases

switch

turn equipment on/off

activate functions - doorbell, alarm, machines

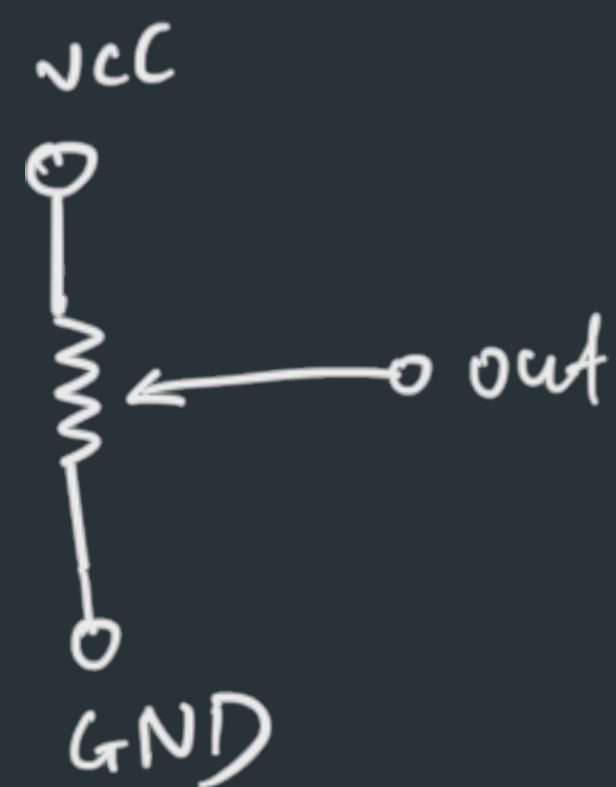
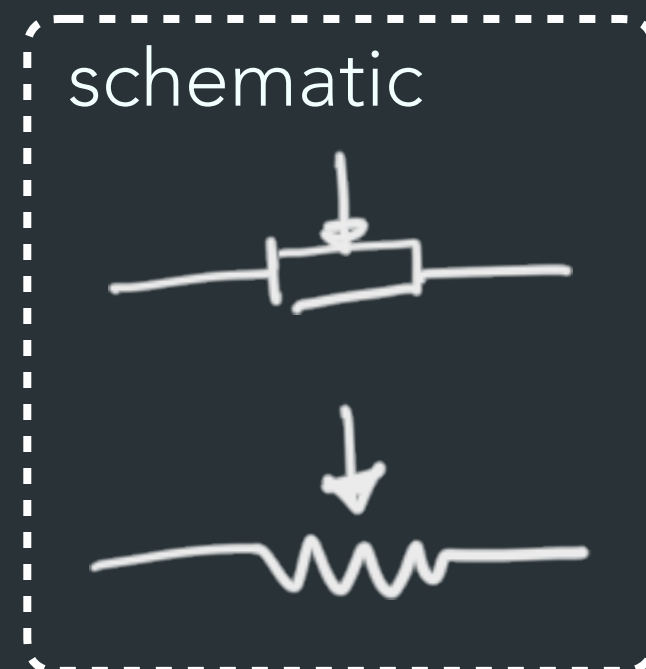
make some sound with **piezo buzzer**



use cases

alarm
tones and melodies
alerts

adjust the resistor with a **potentiometer**



wiper along a
resistive element

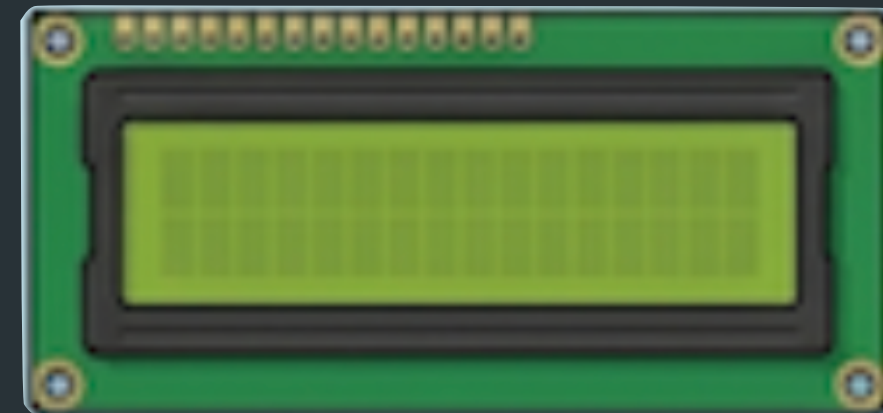
turn knob

adjust
resistance

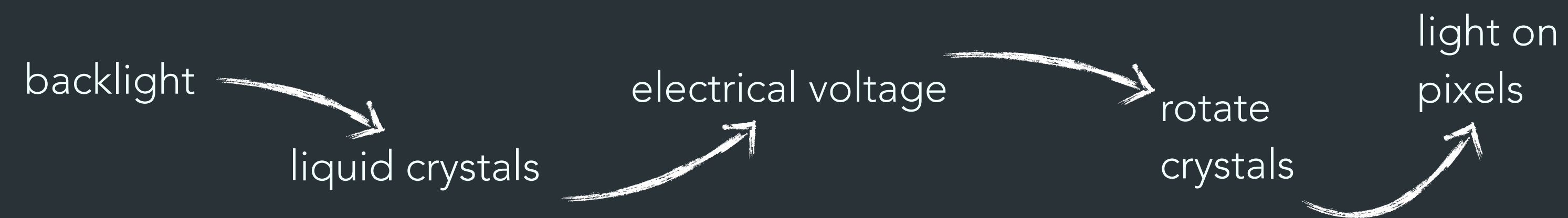
use cases

control motor speed
adjust brightness
control volume

see the outputs on **LCD display**



liquid crystal
display



use cases

digital clock
information board



also nice to have a **multimeter**
to debug, measure and
control electrical quantities

measure

voltage (V)

resistance (Ω)

current (A)